# Model Serving: REST, gRPC, and Streaming

Scaler School of Technology | ML System Design and LLMOps | Week 2 — Session 2/2 | Pre-Lecture Prep

## What Is This Lecture About?

Last session you built a mental model of the production ML system landscape. This session gets specific: how does a model actually receive requests, compute predictions, and return answers — reliably, at scale, with measurable latency guarantees? We move from theory to architecture. By the end of this session, you will be able to choose the right serving mode, reason about latency and throughput trade-offs, and design a deployment strategy that minimises risk.

### From Model to Production Service

The reality of model deployment: companies run hundreds of models, not one or two. We debunk four common myths and confront what degradation, scale, and update cadence actually mean in practice. Source: Chip Huyen, *Designing ML Systems*, Ch. 7.

### Batch vs. Online Prediction Modes

Three prediction modes and when to use each. The DoorDash delivery time example: why you often need *both* historical batch features and real-time streaming features simultaneously. The critical distinction between online features and streaming features.

### Latency, Throughput, and SLOs

Why your mean latency is lying to you. The batching paradox: how accepting slightly higher latency can give you 5× the throughput. Setting SLOs using percentiles (p50, p95, p99). Real-world cost of latency from Akamai and Booking.com studies.

### Serving Patterns and Deployment Strategies

Four serving patterns: synchronous REST, asynchronous batch, streaming, and edge. Three deployment strategies: shadow, canary, and blue-green. A/B testing as a deployment tool. A hands-on look at BentoML and NVIDIA Triton Inference Server.

## What to Expect in the Session

- Opening: how would you serve a model to 1 million users?
- Deployment myths discussion — which ones have you believed?
- Architecture diagrams: three prediction modes side-by-side
- DoorDash example: batch + streaming features in one system
- Latency quiz: diagnosing a p99 latency problem
- Deployment strategy comparison: shadow vs. canary vs. blue-green
- Hands-on look at BentoML (wrapping a model in a REST API)
- Two embedded quizzes and a class discussion

## How to Prepare

- Think about a model you have trained — how would you serve it to 1 million users? What breaks first?
- Recall REST APIs from your software engineering background: what does a synchronous HTTP request look like end-to-end?
- Optional: skim Ch. 7 of Chip Huyen's *Designing ML Systems* ("Model Deployment and Prediction

Service") — focus on the prediction modes and the myths sections
- Optional: look up what gRPC is and how it differs from REST in one paragraph

- Ensure you have Google Colab access — we will look at a BentoML example together
- No new installs needed before class; BentoML will be shown as a demo

**Duration:** ∼2 hours **Prerequisites:** Week 1 Sessions 1 & 2, REST APIs, basic systems thinking **Instructor:** Priyansh Saxena